```
83.149.9.216 - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:43 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboard3.png HTTP/1.1" 200 171717
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:47 +0000] "GET /presentations/logstash-monitorama-2013/plugin/highlight/highlight.js HTTP/1.1" 200 26185
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:12 +0000] "GET /presentations/logstash-monitorama-2013/plugin/zoom-js/zoom.js HTTP/1.1" 200 7697
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:07 +0000] "GET /presentations/logstash-monitorama-2013/plugin/notes/notes.js HTTP/1.1" 200 2892
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:34 +0000] "GET /presentations/logstash-monitorama-2013/images/sad-medic.png HTTP/1.1" 200 430406
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
83.149.9.216 - - [17/May/2015:10:05:57 +0000] "GET /presentations/logstash-monitorama-2013/css/fonts/Roboto-Bold.ttf HTTP/1.1" 200 38720
"http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/32.0.1700.77 Safari/537.36"
```

Figure 3.6 Sample Apache web server log file

Logs are continuously written to the current day's log file. Each log statement has an associated timestamp. At midnight, the file is backed up with a date stamp, and a new file is created for the new day. The log framework initiates the rollover automatically during the day cutover.

If we wish to hold log data in Elasticsearch, we need to rethink the strategy for indexing data that changes/rolls over periodically. We can write an index-rollover script that rolls over the indexes every day at midnight—but there's more to this than just rolling over data. For example, we also need to direct search requests against a single *mother* index rather than multiple rolling indexes. That is, we don't want to issue queries like `GET index1,index2,index3/_search`, specifying individual indexes; instead, we want to invoke `GET myalias/search`, returning data from all underlying indexes. We create an alias for this purpose in chapter 6.

> **DEFINITION** An *alias* is an alternate name for a single index or a set of multiple indexes. The ideal way to search against multiple indexes is to create an alias pointing to them. When we search against an alias, we are essentially searching against all the indexes managed by this alias.

This leads us to an important concept called a *data stream*: an index mechanism to hold time-series data. This is discussed in the next section.

TIME-SERIES DATA

Data streams accommodate time-series data in Elasticsearch—they let us hold data in multiple indexes but allow access as a single resource for search and analytical related queries. As discussed earlier, data tagged to a date or time axis, such as logs, automated car events, daily weather forecasts, pollution levels in a city, Black Friday sales by hour, and so on, is expected to be hosted in timed indexes. On a high level, these indexes are called *data streams*. Behind the scenes, each data stream has a set of indexes for each time point. These indexes are auto-generated by Elasticsearch and are hidden.